

General instructions for running programs.

Since the programs were run in Google Colab, the description is specific to that environment.

1 The first cell mounts the Colab directory for the given program.

For example:

```
from google.colab import drive
drive.mount("/content/drive", force_remount=True)
curr_dir = "/content/drive/MyDrive/KAZAKH_BPE"
%cd "$curr_dir"
```

2 This directory should store the source files for this program. The program's output is also written to this directory.

3 The source data required for the program can be determined from the file paths.

For example:

Paths

```
corpus_path = "/content/drive/MyDrive/KAZAKH_BPE/chunk_099.txt"
spm_model_path = "/content/drive/MyDrive/KAZAKH_BPE/kazakh_bpe.model",
```

The first line specifies the path to the source file, and the second line specifies the path to the program module being used.

4 The program also specifies the import of the necessary packages for running the program.

For example:

```
import math
from collections import Counter
from tqdm import tqdm
```

```
import sentencepiece as spm

from transformers import MT5Tokenizer, AutoTokenizer
```

Sometimes, installing a required library is required to import a required package. For example:

```
!pip install -q pytorch-crf
```

```
import torch

import torch.nn as nn

from torchcrf import CRF

from torch.utils.data import Dataset, DataLoader
```

All of this is written in the program text, written in Python.

5 The program results can be output to a file or directly to the screen.

For example:

```
# Printing results
```

6 Running the program

Program execution is performed according to the rules of the Colab platform. You can execute the entire program or each program cell separately.